AI Smart Study Planner with Stress Prediction

NAME :- Achal Vinod Ghate

Reg no.:- 25BAI11251

Project goal :- To develop a simple AI-based system that predicts a student's stress level based on their daily routine and provides basic study suggestions to help improve time management and reduce stress.

Faculty :- J.MANIKANDAN

1. Introduction

In today's academic environment, students often struggle to maintain a proper balance between study, rest, and personal time. This imbalance can lead to increased stress levels, which directly affects productivity and overall performance.

This project focuses on building a simple system that can analyze a student's daily routine and predict their stress level using Machine Learning. Along with prediction, the system also provides a basic study plan to help improve their routine.

The aim of this project is to apply AI/ML concepts to solve a real-life problem in a simple and practical way.

---

2. Objective

The main objectives of this project are:

- To predict stress level based on daily activities
- To help students understand their routine better
- To provide a simple and useful study plan
- To implement a basic Machine Learning model in a real-world scenario

---

3. Problem Statement

Many students do not follow a structured study routine, which leads to unnecessary stress and reduced efficiency. There is a

need for a system that can analyze daily habits and provide insights in a simple and understandable manner.

This project attempts to solve this problem by predicting stress levels based on input factors like study hours, sleep, screen time, and difficulty level.

---

4. Methodology

The project follows a straightforward approach:

1. A dataset is created with relevant features such as study hours, sleep hours, screen time, and difficulty level.
2. The stress level is defined as the target variable (Low, Medium, High).
3. The data is preprocessed and converted into numerical format.
4. A Machine Learning model (Random Forest Classifier) is trained using this dataset.
5. The trained model is saved and used in the web application.
6. A Flask-based web interface is created where users can input their data.
7. The system predicts the stress level and displays a suitable study plan.

---

5. Tools and Technologies Used

- Python
- Pandas and NumPy

- Scikit-learn
- Flask (for backend)
- HTML and CSS (for frontend)

---

6. System Implementation

a) Model Training

The dataset is used to train a Random Forest model, which performs well for classification problems. The model is evaluated using accuracy score.

study-planner-ai > train_model.py > ...

```python
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score
import pickle
import os

# load dataset
df = pd.read_csv("data/student_stress.csv")

# convert stress labels
df["stress"] = df["stress"].map({
    "Low": 0,
    "Medium": 1,
    "High": 2
})

# features and target
X = df[["study_hours", "sleep_hours", "screen_time", "difficulty"]]
y = df["stress"]

# split data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

# train model
model = RandomForestClassifier()
model.fit(X_train, y_train)

# check accuracy
y_pred = model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)

print(f"Model Accuracy: {accuracy * 100:.2f}%")

# save model
os.makedirs("model", exist_ok=True)
```

```python
with open("model/model.pkl", "wb") as file:
    pickle.dump(model, file)

print("Model trained successfully")
```

b) Web Application

A simple web interface is designed using HTML and CSS. Flask is used to connect the frontend with the Machine Learning model.

```
study-planner-ai > templates > index.html > html
<!DOCTYPE html>
<html lang="en">

<head>
    <meta charset="UTF-8">
    <title>Study Planner</title>

    <style>
        body {
            font-family: Arial, sans-serif;
            background: linear-gradient(to right, #1c1c2b, #2c3e50);
            color: white;
        }

        .container {
            width: 420px;
            margin: 80px auto;
            padding: 25px;
            background-color: rgba(0, 0, 0, 0.6);
            border-radius: 12px;
            text-align: center;
        }

        input {
            width: 85%;
            padding: 10px;
            margin: 8px 0;
            border-radius: 6px;
            border: none;
        }

        button {
            margin-top: 15px;
            padding: 10px 20px;
            background-color: #3cb371;
            border: none;
            border-radius: 8px;
```

```css
input {
    width: 85%;
    padding: 10px;
    margin: 8px 0;
    border-radius: 6px;
    border: none;
}

button {
    margin-top: 15px;
    padding: 10px 20px;
    background-color: #3cb371;
    border: none;
    border-radius: 8px;
    color: white;
    cursor: pointer;
}

button:hover {
    background-color: #2e8b57;
}

.result {
    margin-top: 20px;
    padding: 15px;
    background-color: rgba(255, 255, 255, 0.1);
    border-radius: 8px;
}

.plan {
    margin-top: 10px;
    text-align: left;
    white-space: pre-line;
```

```html
        }
    </style>
</head>

<body>

    <div class="container">
        <h2>Smart Study Planner</h2>
        <p>Enter your routine to check stress level</p>

        <form action="/predict" method="post">
            <input type="number" name="study" placeholder="Study hours" required>
            <input type="number" name="sleep" placeholder="Sleep hours" required>
            <input type="number" name="screen" placeholder="Screen time" required>
            <input type="number" name="difficulty" placeholder="Difficulty (1-10)" required>

            <button type="submit">Check</button>
        </form>

        {% if result %}
        <div class="result">
            <p><b>Stress Level:</b> {{ result }}</p>

            <div class="plan">
                <b>Study Plan:</b> {{ plan }}
            </div>
        </div>
        {% endif %} {% if show_button %}
        <form action="/" method="get">
            <button type="submit">Check Again</button>
        </form>
        {% endif %}
```

```
    </div>

</body>

</html>
```

c) User Interaction

The user enters:

- Study hours
- Sleep hours
- Screen time
- Difficulty level

Based on this input, the system predicts stress level and suggests a study plan.

---

7. Features of the System

- Simple and user-friendly interface
- Real-time stress prediction
- Personalized study plan
- “Check Again” feature for multiple inputs
- Backend logging of user inputs and results

---

8. Results

The model successfully predicts stress levels based on user input. The accuracy of the model is displayed during training. The system provides reasonable suggestions that can help improve study habits.

---

9. Applications

- Can be used by students for self-analysis
- Useful for basic academic planning
- Can be extended into a full productivity tracking system

---

10. Limitations

- Dataset is small and manually created
- Predictions are based on limited parameters
- Not suitable for complex psychological analysis

---

11. Future Scope

- Use a larger and real-world dataset
- Improve model accuracy with advanced algorithms
- Add user login and history tracking
- Convert into a mobile or web-based platform

---

12. Conclusion

This project demonstrates how Machine Learning can be used to solve a common student problem in a simple way. It not only predicts stress levels but also encourages better study habits through basic planning.

Even though the system is simple, it provides a strong foundation for building more advanced applications in the future.

---

13. References

- Scikit-learn documentation
- Python official documentation
- Online tutorials and resources for Flask and Machine Learning

---

14. Author

Developed by: Achal ghate
Course: Fundamental of AI & ML